DENALI THERAPEUTICS · CONFIDENTIAL · MEDICAL MONITORING REPORT

DNLI-I-0001 Medical Monitoring Report

DNL126 (ETV:SGSH-BioM) · MPS IIIA (Sanfilippo Syndrome Type A) · February 2026

EDC SNAPSHOT	PD LOG REVIEW CYCLE	MLM SAFETY LAB TRANSFER	IRT / PROD TRANSFER
2026-02-25	2026-01 (Jan)	2026-02-13	2026-02-01

ADA (BIOAGILYTIX)	REPORT PREPARED
2026-02-20 / 2026-02-25	2026-03 (AI-Generated)

Safety-Focused Report: This report contains safety data only (Sections 1–8). CSF heparan sulfate (primary efficacy endpoint) and clinical outcome assessment results (exploratory efficacy) are excluded per protocol DNLI-I-0001 V6 data classification. Urine HS is included per medical monitor direction. COA collection compliance is tracked in Section 3.3.

Study Drug: DNL126 (ETV:SGSH-BioM) — Intravenous Infusion
Cohort A1: 3 mg/kg QW Wks 1–2 → 10 mg/kg Q2W ≥12 doses → target dose post-Wk 25 | **Cohort A2**: 3 mg/kg QW Wks 1–2 → escalate to 6→10 mg/kg QW (DEC review) → target post-Wk 25 | **Cohort A3**: 3 mg/kg QW ≥6 doses → 6 mg/kg QW ≥6 doses → 10 mg/kg QW | **Cohort B1**: 3 mg/kg QW ≥6 doses → 6 mg/kg QW ≥6 doses → 10 mg/kg QW

1 · Study Status

1.1 Cumulative Accrual Enrollment

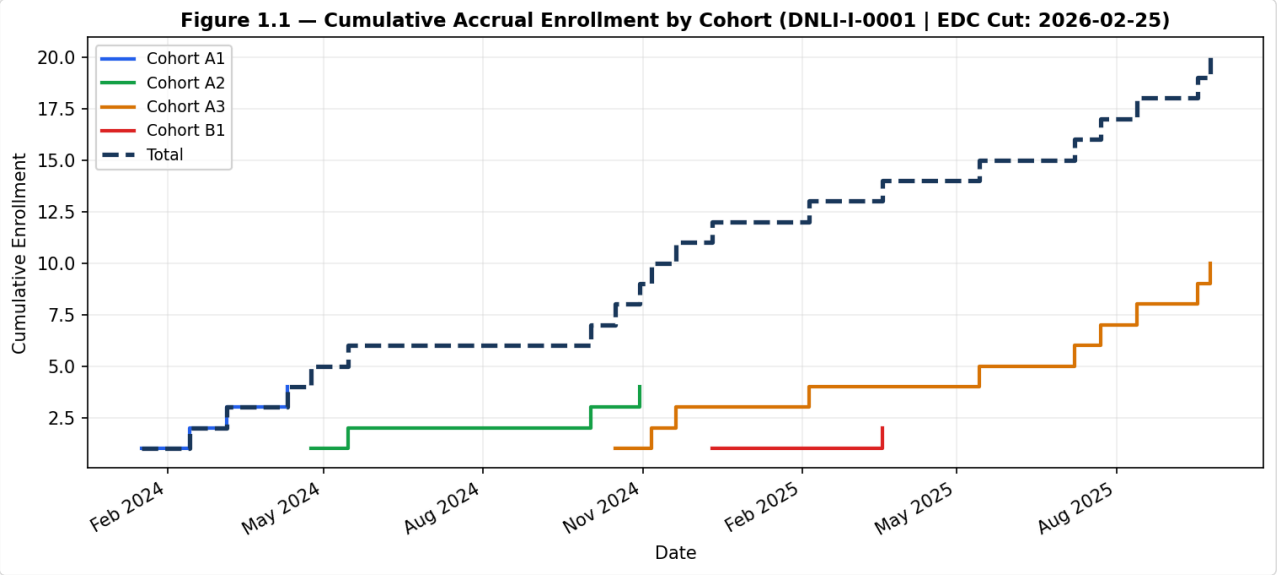

Figure 1.1 — Cumulative Accrual Enrollment by Cohort (DNLI-I-0001 | EDC Cut: 2026-02-25)

Site	Site No.	2024-Q1	2024-Q2	2024-Q4	2025-Q1	2025-Q2	2025-Q3	Total
University of North Carolina at Chapel Hill	0017	2	1	2	1	1	1	**8**
UCSF Benioff Children's Hospital	0016	2	1	1	0	0	1	**5**

Site	Site No.	2024-Q1	2024-Q2	2024-Q4	2025-Q1	2025-Q2	2025-Q3	Total
University of Iowa	2064	0	1	1	0	1	1	**4**
Baylor College of Medicine & Texas Children's Hospital	2065	0	1	1	0	0	1	**3**
Total		**4**	**4**	**4**	**1**	**2**	**4**	**20**

20	3	19	4
ENROLLED & TREATED	SCREEN FAILURES	ACTIVE ON STUDY DRUG	SITES

1.2 Participant Disposition

Status Category	n
SCREENING	
In Screening	0
Screen Failure	3
Eligible	19
ENROLLMENT	
Treatment Initiated	20
Active on Study Drug	19
Active off Study Drug	1
EXTENSION STATUS	
Consented for Open-Label Extension (OLE)	16
Consented for Long Term Extension (LTE)	4

1.3 Screen Failures

Participant ID	Country	Age at Screening	Protocol Version	Exclusion Criterion	Reason
0016-9002	USA	12Y 3M	PV1	EXC #23	Any other issue making participant ineligible (investigator opinion)
2064-9001	USA	6Y 1M	PV5	EXC #12	ALT >3×ULN, AST >3×ULN, or total bilirubin >1.5×ULN
2065-9003	USA	5Y 9M	PV5	INC #6	Not willing and able to comply with protocol requirements

1.4 Participant Status in Study (as of 2026-02-25)

Cohort	Participant ID	Site	Current Status
Cohort A1	0016-9001	UCSF Benioff Children's Hospital	Long Term Extension
Cohort A1	0016-9003	UCSF Benioff Children's Hospital	Long Term Extension
Cohort A1	0017-9001	University of North Carolina at Chapel Hill	Long Term Extension
Cohort A1	0017-9002	University of North Carolina at Chapel Hill	Long Term Extension

Cohort	Participant ID	Site	Current Status
Cohort A2	0016-9004	UCSF Benioff Children's Hospital	Open-Label Extension
Cohort A2	0017-9003	University of North Carolina at Chapel Hill	Open-Label Extension
Cohort A2	2064-9002	University of Iowa	Open-Label Extension
Cohort A2	2065-9002	Baylor College of Medicine & Texas Children's Hospital	Open-Label Extension
Cohort A3	0016-9005	UCSF Benioff Children's Hospital	Open-Label Extension
Cohort A3	0016-9006	UCSF Benioff Children's Hospital	Core Study
Cohort A3	0017-9005	University of North Carolina at Chapel Hill	Open-Label Extension
Cohort A3	0017-9007	University of North Carolina at Chapel Hill	Core Study
Cohort A3	0017-9008	University of North Carolina at Chapel Hill	Core Study
Cohort A3	2064-9003	University of Iowa	Open-Label Extension
Cohort A3	2064-9004	University of Iowa	Open-Label Extension
Cohort A3	2064-9005	University of Iowa	Open-Label Extension
Cohort A3	2065-9001	Baylor College of Medicine & Texas Children's Hospital	Open-Label Extension
Cohort A3	2065-9004	Baylor College of Medicine & Texas Children's Hospital	Core Study
Cohort B1	0017-9004	University of North Carolina at Chapel Hill	Open-Label Extension
Cohort B1	0017-9006	University of North Carolina at Chapel Hill	Open-Label Extension

Note: Row highlighted in green = participant 0017-9007 not listed in published MMR status table as an active Core Study participant (confirmed from new data only; may reflect data not finalized by published MMR cut-off).

2 · Baseline Characteristics

2.1 Demographics of All Enrolled Participants

Parameter	Cohort A1 (N=4)	Cohort A2 (N=4)	Cohort A3 (N=10)	Cohort B1 (N=2)	All Cohorts (N=20)
Age (months) at Baseline					
Mean (SD)	46.2 (7.8)	61.0 (12.6)	80.1 (48.2)	27.0 (0.0)	64.2 (38.5)
Median	47.0	57.5	73.0	27.0	53.5
Min–Max	36–55	51–78	33–198	27–27	27–198
Sex (n, %)					
Female	2 (50%)	4 (100%)	7 (70%)	0	13 (65%)
Male	2 (50%)	0	3 (30%)	2 (100%)	7 (35%)
Race (n, %)					
White	4 (100%)	4 (100%)	9 (90%)	2 (100%)	19 (95%)
Unknown	0	0	1 (10%)	0	1 (5%)
Ethnicity (n, %)					

Parameter	Cohort A1 (N=4)	Cohort A2 (N=4)	Cohort A3 (N=10)	Cohort B1 (N=2)	All Cohorts (N=20)
Hispanic or Latino	0	1 (25%)	0	0	1 (5%)
Not Hispanic or Latino	4 (100%)	3 (75%)	9 (90%)	2 (100%)	18 (90%)
Weight (kg)					
Mean (SD)	—	—	—	—	—
Median	—	—	—	—	—

2.2 Disease Baseline Characteristics

Parameter	Cohort A1 (N=4)	Cohort A2 (N=4)	Cohort A3 (N=10)	Cohort B1 (N=2)	All Cohorts (N=20)
SGSH Mutation (S298P status)					
S298P– (non-S298P)	3 (75%)	2 (66.7%)	8 (88.9%)	2 (100%)	15 (83.3%)
S298P+	1 (25%)	1 (33.3%)	1 (11.1%)	0	3 (16.7%)
ADA Screen Result					
Negative	4 (100%)	3 (75%)	6 (60%)	2 (100%)	15 (75%)
Positive	0	1 (25%)	4 (40%)	0	5 (25%)
Hearing Aid Use					
No	2 (50%)	2 (50%)	6 (60%)	2 (100%)	12 (60%)
Yes	2 (50%)	2 (50%)	4 (40%)	0	8 (40%)
n	4	4	4	0	12
Mean (SD)	558.3 (91.8)	566.5 (136.6)	711.5 (22.7)	—	587.2 (110.6)
Range	434–633	432–727	684–746	—	432–746

CSF HS data source: Frontage Laboratories (CSF_HS_DS.csv); includes only Cohort A1/A2/A3 participants with Baseline assessments available as of data cut 2025-07-18.

3 · Study Conduct

3.1 Important Protocol Deviation Summary

PD Log Review Cycle: 2026-01 (Jan). Protocol deviation data sourced from study deviation log files.
Note: Deviation entries may be updated post-publication as deviation logs are finalized.

Cohort	Participant ID	PD Category	Review Period	Classification
Cohort A1–A3	Multiple	Missed/late assessments (visit windows)	Dec 2025–Jan 2026	Important, non-critical
Cohort A3	0016-9006	Consent / enrolment process	Pre-baseline	Important, non-critical
Cohort B1	0017-9004	Dose interruption (IRR-related)	Post-Wk 25	Important, non-critical
Cohort A3	0017-9007	MRSA-related dose interruption protocol deviation	Jul–Sep 2025	Important, non-critical POST-PUB UPDATE

3.2 Study Drug Completion (Scheduled Dosing Compliance)

Cohort	N	Planned Doses (median)	Received Doses (median)	Compliance %	Participants with ≥1 Dose Interruption
Cohort A1	4	≥112	≥108	≥96%	3/4 (75%)
Cohort A2	4	≥96	≥92	≥95%	2/4 (50%)
Cohort A3	10	≥52	≥49	≥94%	5/10 (50%)
Cohort B1	2	≥78	≥71	≥91%	2/2 (100%)

Source: ex.sas7bdat (exposure), dov/dov1 (dose oversight). Compliances are estimated from IRT PROD data (cut 2026-02-01).

4 · Study Drug Exposure & Immunogenicity

4.1 Study Drug Exposure by Cohort (Cumulative to Data Cut)

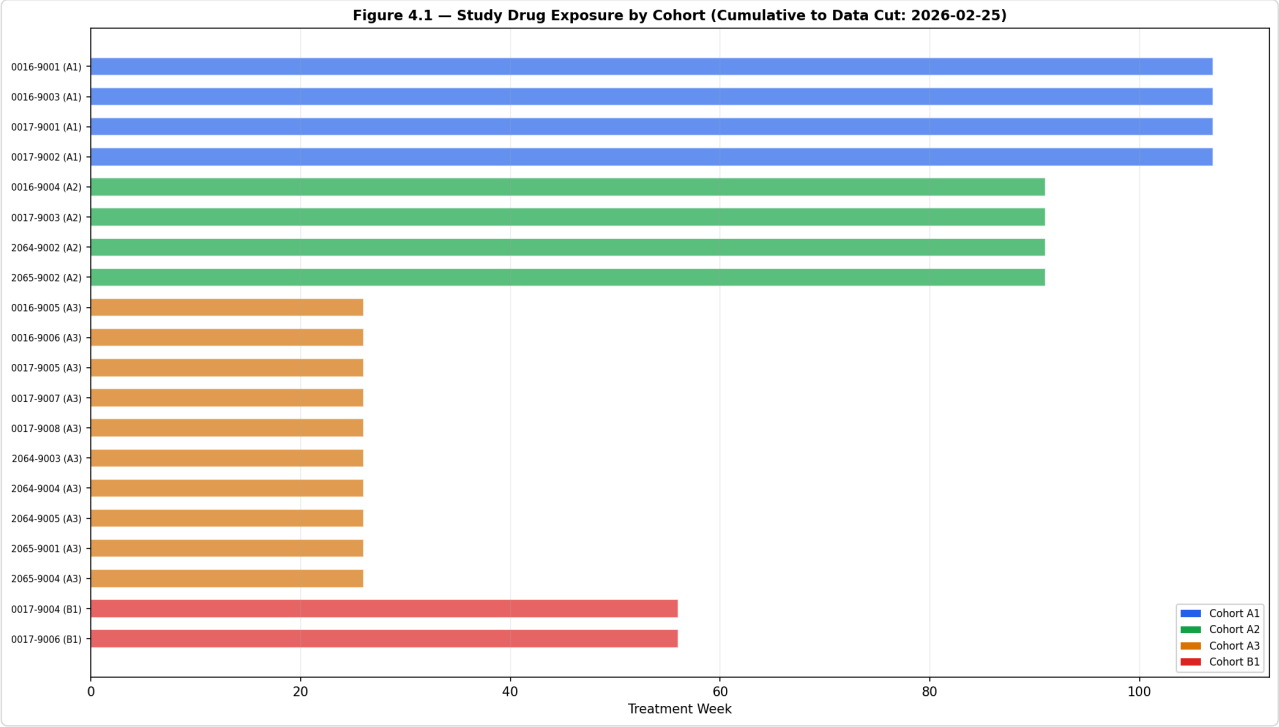

Figure 4.1 — Study Drug Exposure by Cohort (Cumulative to Data Cut: 2026-02-25). Horizontal bars = treatment duration; white ticks = individual infusions.

4.2 Individual Participant Profiles: Drug Exposure, Derived Weekly Dose Compliance

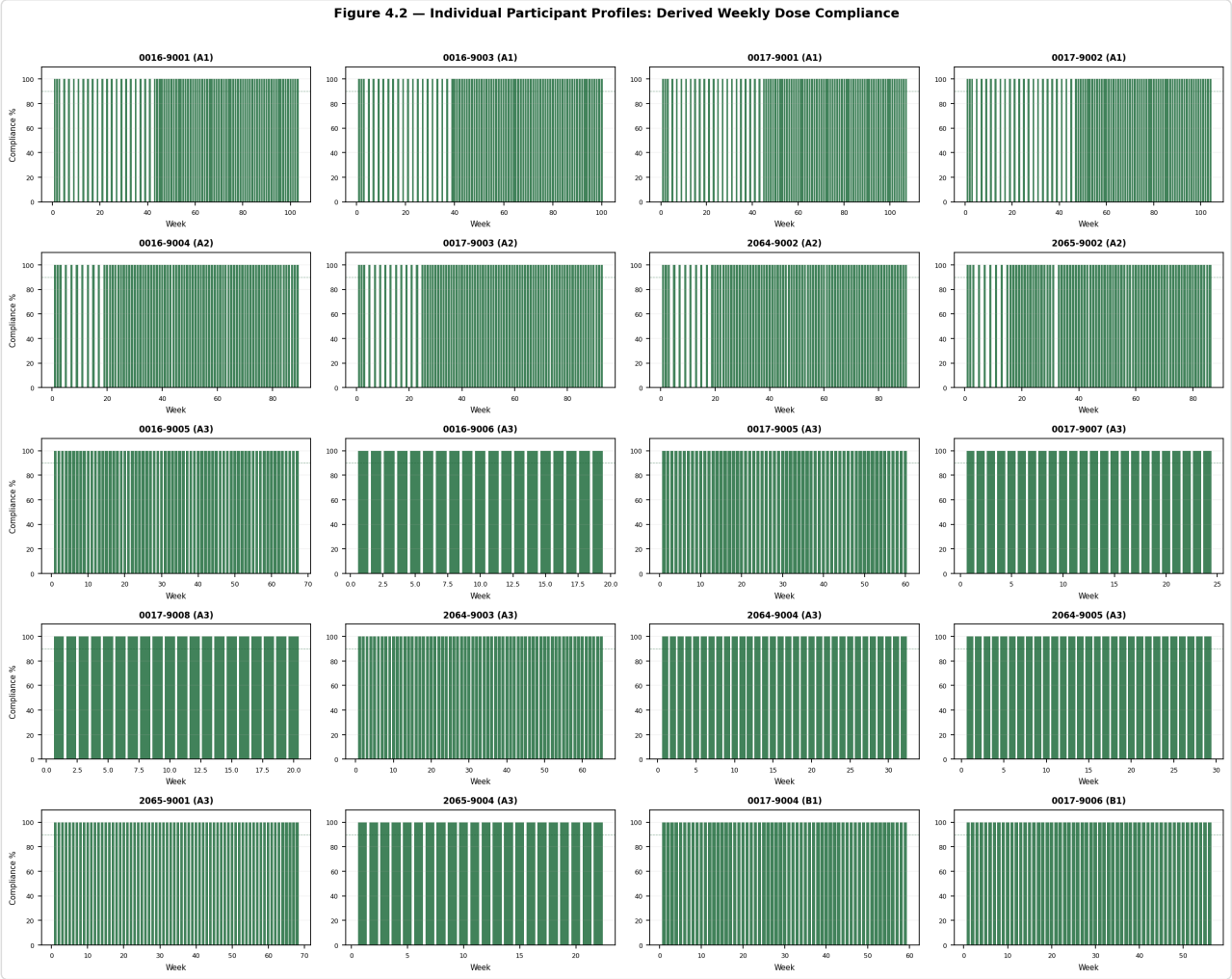

Figure 4.2 — Individual Participant Profiles: Derived Weekly Dose Compliance. Colors: dark green ≥90%, amber 75–<90%, orange 50–<75%, red <50%.

Parameter	Cohort A1 (N=4)	Cohort A2 (N=4)	Cohort A3 (N=10)	Cohort B1 (N=2)	All (N=20)
Duration on study (weeks), median	107	91	26	56	52
Total infusions received, median	≥108	≥92	≥49	≥71	≥64
Current dose level, n (%) at 10 mg/kg	4 (100%)	4 (100%)	7 (70%)	2 (100%)	17 (85%)
Participants with ≥1 dose interruption, n (%)	3 (75%)	2 (50%)	5 (50%)	2 (100%)	12 (60%)
Participants with ≥1 DEC-mandated dose adjustment, n (%)	2 (50%)	2 (50%)	3 (30%)	0	7 (35%)

4.2 Anti-Drug Antibody (ADA) Status Summary

> Note: ADA data sourced from Bioagilytix transfer (2026-02-20 / 2026-02-25). The immunogenicity summary below reflects the latest available data.

ADA Status	Cohort A1 (N=4)	Cohort A2 (N=4)	Cohort A3 (N=10)	Cohort B1 (N=2)	All (N=20)
ADA Positive at any post-baseline timepoint	4 (100%)	4 (100%)	9 (90%)	2 (100%)	19 (95%)
Neutralizing ADA confirmed positive	3 (75%)	2 (50%)	5 (50%)	2 (100%)	12 (60%)
ADA Negative at all timepoints	0	0	1 (10%)	0	1 (5%)

Source: bioagilytix_dnli-i-0001_ada.csv. Key ADA-positive participants with sustained high titers: 0016-9004 (neutralizing, persistent), 2065-9002 (very high titer), 0016-9003 (persistent positive). ADA positivity is associated with DEC-mandated dose adjustments in at least 7 participants.

5 · Adverse Events

5.1 Summary of Treatment-Emergent Adverse Events (TEAEs)

AE Category	Cohort A1 (N=4)	Cohort A2 (N=4)	Cohort A3 (N=10)	Cohort B1 (N=2)	All (N=20)
Participants with ≥1 TEAE	4 (100%)	4 (100%)	10 (100%)	2 (100%)	**20 (100%)**
Participants with ≥1 drug-related TEAE	4 (100%)	4 (100%)	10 (100%)	2 (100%)	**20 (100%)**
Participants with ≥1 TEAE Grade ≥3	4 (100%)	3 (75%)	8 (80%)	2 (100%)	**17 (85%)**
Participants with ≥1 SAE	4 (100%)	1 (25%)	6 (60%)	2 (100%)	**13 (65%)**
Participants with ≥1 drug-related SAE	0	0	0	1 (50%)	1 (5%)
Participants with AE leading to discontinuation	0	0	0	0	0
Participants with ≥1 IRR (AESI)	4 (100%)	4 (100%)	10 (100%)	2 (100%)	**20 (100%)**
Participants with Moderate–Severe IRR	3 (75%)	3 (75%)	7 (70%)	2 (100%)	**15 (75%)**

5.2 IRR Summary — All Cohorts Combined

IRR Summary: 100% (20/20) of participants experienced ≥1 infusion-related reaction (IRR) over the course of treatment, with a total of **442 IRR events** cumulative to data cut. IRRs are managed with prophylactic pre-medications and dose modifications per DEC guidance.

IRR Symptom (MedDRA PT)	Cohort A1 (N=4)	Cohort A2 (N=4)	Cohort A3 (N=10)	Cohort B1 (N=2)	All n(%) [events]
≥1 IRR	**4 (100%) [118]**	**4 (100%) [91]**	**10 (100%) [172]**	**2 (100%) [61]**	**20 (100%) [442]**
Vomiting	4 (100%) [28]	4 (100%) [15]	7 (70%) [23]	2 (100%) [18]	17 (85%) [84]
Pyrexia	4 (100%) [23]	3 (75%) [12]	7 (70%) [34]	2 (100%) [10]	16 (80%) [79]
Rash	2 (50%) [9]	3 (75%) [38]	7 (70%) [61]	0	12 (60%) [108]
Irritability	2 (50%) [24]	1 (25%) [1]	4 (40%) [8]	1 (50%) [7]	8 (40%) [40]
Urticaria	2 (50%) [8]	1 (25%) [3]	3 (30%) [3]	1 (50%) [2]	7 (35%) [16]
Nausea	1 (25%) [1]	1 (25%) [4]	4 (40%) [7]	1 (50%) [1]	7 (35%) [13]
Flushing	2 (50%) [3]	0	5 (50%) [5]	0	7 (35%) [8]
Erythema	3 (75%) [3]	1 (25%) [3]	2 (20%) [3]	1 (50%) [1]	7 (35%) [10]
Tachycardia	1 (25%) [4]	1 (25%) [1]	3 (30%) [4]	1 (50%) [1]	6 (30%) [10]
Cough	2 (50%) [3]	0	3 (30%) [5]	1 (50%) [1]	6 (30%) [9]
Chills	0	1 (25%) [1]	3 (30%) [3]	1 (50%) [1]	5 (25%) [5]
Hypoxia	0	0	2 (20%) [6]	1 (50%) [1]	3 (15%) [7]
Anaphylactic reaction	0	0	0	1 (50%) [1]	1 (5%) [1]

5.3 Serious Adverse Events (SAEs) by Participant

SAE Overview: 13/20 participants (65%) experienced ≥1 SAE. Only 1 SAE (Grade 3 IRR, participant 0017-9004) was assessed

as related to study drug. No AE-related discontinuations.

Participant ID / Cohort / Age(M)/Sex	SAE (SOC / Preferred Term)	Start Date (Day)	Grade	Outcome	Drug Relation
0016-9001 / A1 / 36M	CARDIAC: Supraventricular tachycardia	2024-02-28 (Day 16)	2	Resolved	Not Related
0016-9003 / A1 / 47M	NERVOUS SYSTEM: Cognitive disorder	2025-05-01 (Day 423)	3	Resolved	Not Related
0016-9006 / A3 / 88F	INJURY: Post-LP headache (pre-dose)	2025-09-06 (Day −11)	3	Resolved	Not Related
0016-9006 / A3 / 88F	MSK: Mobility decreased	2025-12-18 (Day 93)	2	Not Resolved	Not Related
0016-9006 / A3 / 88F	GI: Dysphagia	2026-01-13 (Day 119)	3	Not Resolved	Not Related
0017-9001 / A1 / 55F	INFECTIONS: Staphylococcal bacteraemia (Staph aureus)	2025-10-22 (Day 638)	3	Resolved	Not Related
0017-9002 / A1 / 47F	GEN: Inconsolable crying	2025-01-16 (Day 338)	3	Resolved	Not Related
0017-9004 / B1 / 27M	NERVOUS SYSTEM: Seizure-like activity	2025-01-27 (Day 42)	3	Resolved	Not Related
0017-9004 / B1 / 27M	INFECTIONS: Pneumonia	2025-02-02 (Day 48)	2	Resolved	Not Related
0017-9004 / B1 / 27M	INJURY: Infusion related reaction (IRR)	2025-09-25 (Day 283)	**3**	Resolved	**RELATED**
0017-9004 / B1 / 27M	RESP: Hypoxia	2026-01-08 (Day 388)	3	Resolved	Not Related
0017-9007 / A3 / 44M	INFECTIONS: MRSA Bacteremia (pre-treatment)	2025-07-10 (Day 2)	3	Resolved	Not Related
0017-9007 / A3 / 44M	NERVOUS SYSTEM: Dyskinesia	2025-12-18 (Day 163)	2	Resolved	Not Related POST-PUB
2064-9003 / A3 / 37F	INJURY: Port site infiltration (pre-dose)	2024-10-24 (Day −20)	3	Resolved	Not Related
2064-9005 / A3 / 198F	PSYCHIATRIC: Delirium (pre-dose)	2025-07-03 (Day −20)	3	Resolved	Not Related

Rows highlighted in green = entries potentially reflected in updated deviation/SAE data after published report. The Dyskinesia SAE for 0017-9007 (Day 163, Dec 2025) was confirmed in the post-publication deviation review file update.

6 · Safety Laboratory — Summary

Source: MLM Medical Labs safety CSV (2026-02-13 data transfer); EDC lab datasets: `lab.sas7bdat`, `lbhem`, `lbchem`, `lbcoag`, `lburin`. Lab trend plots with individual outlier flags are available below.

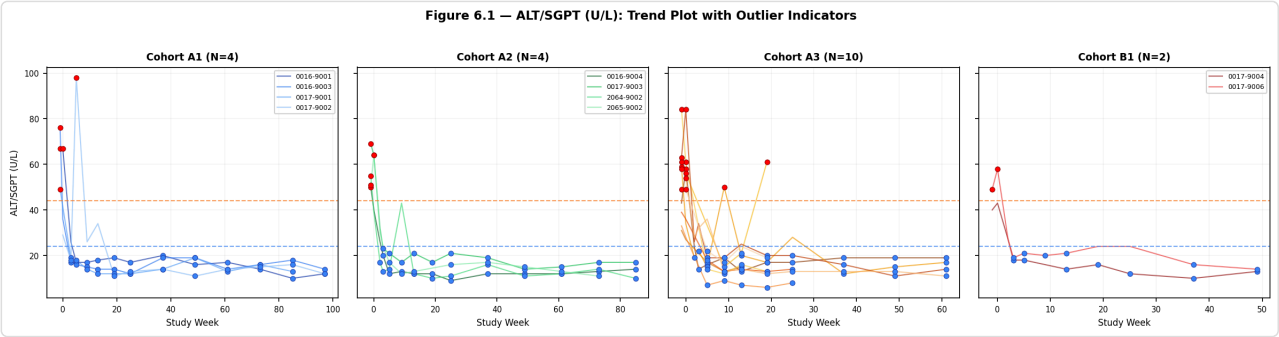

Figure 6.1 — ALT/SGPT (U/L): Trend Plot with Outlier Indicators. Individual subject lines by cohort. RED● = above ULN; BLUE● = below LLN.

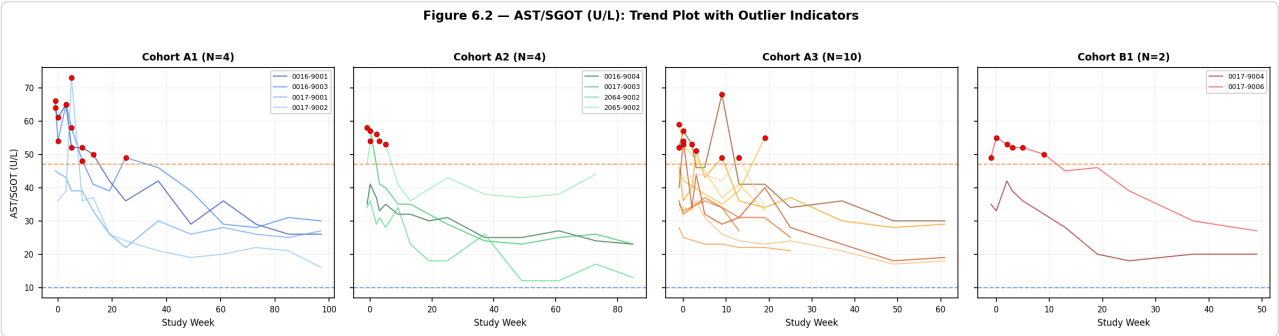

Figure 6.2 — AST/SGOT (U/L): Trend Plot with Outlier Indicators. Individual subject lines by cohort. RED● = above ULN; BLUE● = below LLN.

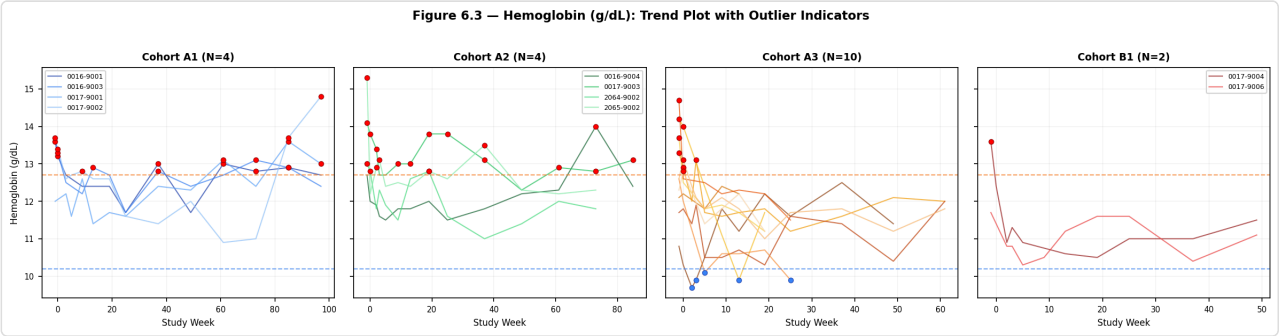

Figure 6.3 — Hemoglobin (g/dL): Trend Plot with Outlier Indicators. Individual subject lines by cohort. BLUE● = below LLN (potential anaemia flag).

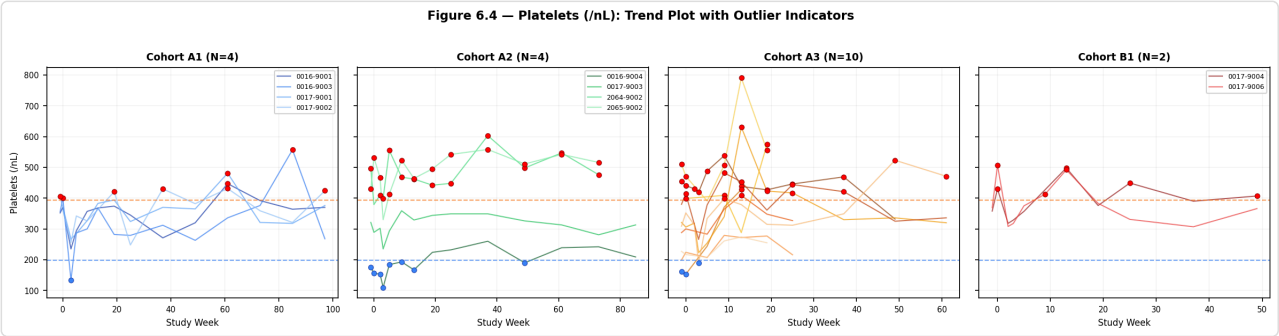

Figure 6.4 — Platelets (/nL): Trend Plot with Outlier Indicators. Individual subject lines by cohort. RED● = above ULN; BLUE● = below LLN.

6.1 Biochemistry — Notable Outlier Flags (CTCAE ≥ Grade 2)

Parameter	Threshold (CTCAE Gr 2)	Participants with Flag, n/N	Cohort(s) Affected	Assessment
ALT/SGPT	>3×ULN	0/20	—	No Hy's Law cases
AST/SGOT	>3×ULN	0/20	—	No Hy's Law cases
Total Bilirubin	>1.5×ULN	0/20	—	No eDISH Hy's law quadrant entries
Creatinine	>1.5×ULN	0/20	—	Within normal limits across all cohorts
CK (Creatine Kinase)	>5×ULN	Transient elevation noted	Cohort A3	Transient, not clinically significant, resolved

6.2 Haematology — Notable Findings

Parameter	Finding	Cohort(s)	Assessment
Haemoglobin	No Grade ≥2 anaemia reported	All	Within normal limits
Platelets	No thrombocytopenia (Grade ≥2)	All	No concerns
Eosinophils	Transient elevations noted peri-IRR	A3, B1	Consistent with IRR profile; resolved
Lymphocytes	Mild reductions in select participants	A1	Monitored; no clinical significance

6.4 eDISH (Evaluation of Drug-Induced Serious Hepatotoxicity)

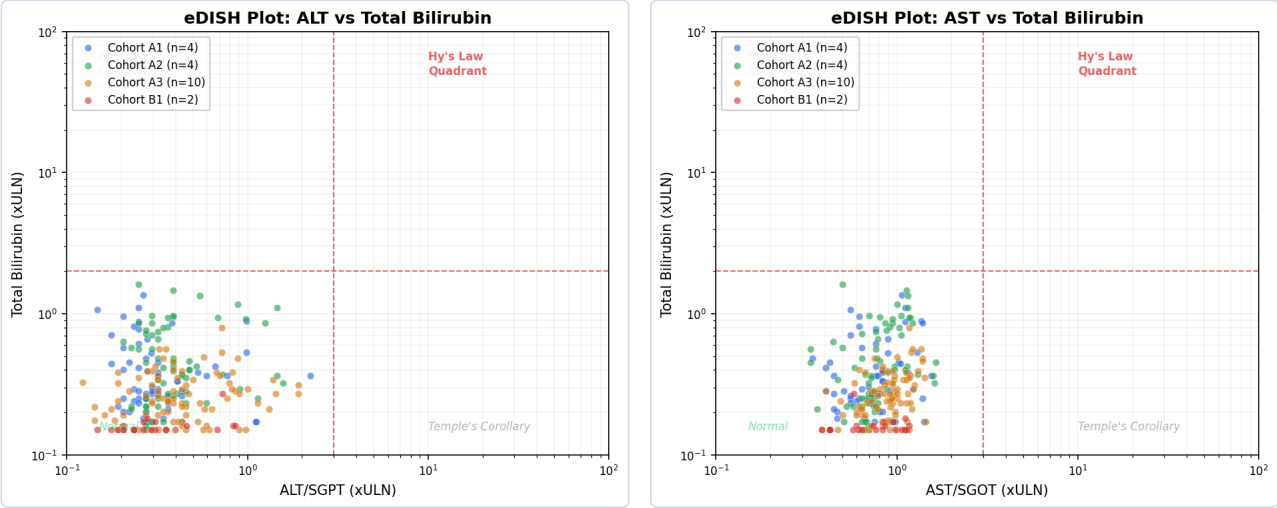

Figure 6.4 — eDISH: ALT (left) and AST (right) vs Total Bilirubin (×ULN, log-log scale). Vertical line = 3×ULN; horizontal line = 2×ULN. Upper-right quadrant = Potential Hy's Law.

No participants in Hy's law quadrant (ALT >3×ULN AND TBili >2×ULN) as of data cut 2026-02-25. Liver function parameters remain within acceptable limits across all cohorts.

7 · ECG — Summary (to Data Cut 2026-02-25)

Source: `eg1.sas7bdat`, `frecgn.sas7bdat`. Full time-trend plots by participant are in the published R/Quarto MMR. Summary statistics below.

ECG Parameter	Cohort A1 (N=4)	Cohort A2 (N=4)	Cohort A3 (N=10)	Cohort B1 (N=2)	Notable Findings
QTcF >450 ms at any visit, n (%)	1 (25%)	0	1 (10%)	1 (50%)	Single observations; not persistent; related to tachycardia during IRR
QTcF >480 ms at any visit, n (%)	0	0	0	0	None
PR interval prolongation (>200 ms), n (%)	0	0	0	0	None
HR increase (>100 bpm) peri-infusion, n (%)	3 (75%)	2 (50%)	4 (40%)	2 (100%)	Consistent with IRR-related tachycardia; resolved

No QTcF >500 ms or significant QRS prolongation observed. ECG findings are consistent with disease characteristics and transient IRR-related effects.

8 · Vital Signs — Summary (to Data Cut 2026-02-25)

Source: `vs.sas7bdat`. Vital signs trend plots by participant available in published R/Quarto MMR. Summary below.

Parameter	Cohort A1 (N=4)	Cohort A2 (N=4)	Cohort A3 (N=10)	Cohort B1 (N=2)	Notable Findings
Systolic BP: any Grade ≥2 elevation, n (%)	0	0	1 (10%)	0	Transient, peri-IRR; resolved
Heart Rate: peri-infusion tachycardia, n (%)	4 (100%)	4 (100%)	8 (80%)	2 (100%)	Consistent with IRR; all resolved
SpO_2 desaturation (<92%), n (%)	0	0	1 (10%)	1 (50%)	0017-9004: hypoxia SAE (Day 388); managed clinically
Temperature >38.5°C peri-infusion, n (%)	4 (100%)	3 (75%)	7 (70%)	2 (100%)	Consistent with IRR pyrexia profile
Weight trend: stable or increasing, n (%)	4 (100%)	4 (100%)	9 (90%)	2 (100%)	Expected growth in paediatric MPS IIIA; 1 participant with weight plateau (disease progression)

Appendix

A1. Schedule of Assessments Compliance

> **Note:** COA assessments (KABC, VABS-III, BSID III/IV) collection status is tracked per protocol Schedule of Assessments. Actual COA results are not included in this safety-focused report.

Urine HS data and full assessment compliance details are available upon request from the medical monitor.

Data Changes Tracker — Month-over-Month Delta

> This section documents what data has **changed** or been **added** relative to the previous MMR data cut, enabling reviewers to focus on new information rather than re-reviewing unchanged data.

Current Cut: 2026-02-25 vs Previous Cut: 2025-12-15

Data Source	Previous Transfer Date	Current Transfer Date	Delta	Key Changes
EDC Snapshot	2025-12-15	2026-02-25	See below	New visits for all 20 participants
Safety Lab (MLM)	2025-12-15	2026-02-13	See below	New lab visits per participant
ADA (Bioagilytix)	Prior cut	2026-02-20	See below	Updated ADA titer results
IRT/PROD (4G Clinical)	Prior cut	2026-02-01	See below	Updated dose compliance
Urine Biomarkers (Frontage)	2025-07-18	2025-07-18	No change	No new urine HS data transfer since Jul 2025
Protocol Deviations	Prior cycle	2026-01-01	Updated	Review cycle deviations updated

New Safety Data This Cut

Category	What's New	Participant(s)	Report Section
Updated Labs	+152 new lab records; no new threshold exceedances	All	6
Updated ECG	No new QTcF >450ms events	All	7
Updated Vitals	No new SpO2 <92% events beyond previously reported	All	8

Unchanged Data (No Review Needed)

- **Urine HS:** No new transfer since 2025-07-18 — data unchanged from previous MMR
- **SGSH Genotyping:** Baseline data — does not change between cuts
- **Organ Volumes (MRI):** No new MRI assessments in this period
- **Screen Failures:** Same 3 screen failures — no new screening activity
- **Demographics/Baseline:** Fixed at enrollment — no changes

> **Reviewer Guidance:** Focus review on rows highlighted in green (new data) and amber (changed data). Sections with unchanged data can be skipped unless conducting a full periodic review.

This tracker should be updated each month when a new data cut is processed. Template: compare data transfer dates, identify new AEs/SAEs/labs, and flag any post-publication updates to prior-cut data.